

Q.1

1. What is Probability?

→ Probability is a measure of how likely an event is to occur. Probability is a tool that converts uncertainty into numbers.

formula $P(A) = \frac{\text{Number of favorable Outcome}}{\text{Number of Total Outcome}}$

Example when rolling a fair dice probability of getting 6...

→ $P(6) = \frac{1}{6} = 0.17$

2. key terminology of Probability

→ i. Experiment = An operation or process that can be repeated and lead to well-defined results

-> Ex = Tossing a coin or rolling a dice

ii. Sample Space (S) = The set of all possible outcomes of a random experiment

→ Ex: for a coin toss, the Sample Space is $S = \{\text{Head, Tail}\}$

iii. Event (E) = A specific outcome or a subset of the sample space that you are looking for

Ex: If you want to roll an even number

2. How at least 3 event from dataset

i. $P(\text{Pass}) \approx 45 \rightarrow P(\text{Pass}) = 0.45$

ii. $P(\text{Group discussion} = \text{yes}) \rightarrow P(G.D = \text{yes}) = 0.49$

iii. $P(\text{Attendance} \geq 80\%)$
 $P(A+ \geq 80\%) = 0.41$

Q.2

1. Identify one Empirical probability and one theoretical probability scenario

$\rightarrow$ Empirical $P(\text{Pass}) = \frac{45}{100}$

$P(\text{Pass}) = 0.45$

Theoretical $P(\text{Select specific student}) = 1/100 = 0.01$

$\rightarrow$ what we call as Theoretical probability the event whose out put is recognizable without an experiment

$\rightarrow$ Some universal Example one Tossing a coin so out come are $\{H, T\}$

$\rightarrow$ but from out our dataset that is $\{Pass, Fail\}$

Q.2 Random Variable and Probability Distribution

$$\rightarrow X = 0, 1, 2, 3$$

x	$P(x)$	$x \cdot P(x)$
0	0.027	0.00
1	0.189	0.189
2	0.441	0.882
3	0.343	1.029
		<u>2.1</u>

$$E[X] = \sum (x \cdot P(x))$$

$$= 0.189 + 0.882 + 1.029 = 2.1$$

$$X = 0, 1, 2, 3$$

x	$(x-\mu)$	$(x-\mu)^2$	$P(x)$	$(x-\mu)^2 \cdot P(x)$
0	-2.1	4.41	0.027	0.119
1	-1.1	1.21	0.189	0.229
2	-0.1	0.01	0.441	0.004
3	0.9	0.81	0.343	0.278

$$\text{Var}(x) = \sum (x_i - \mu)^2 \cdot P(x_i)$$

$$= 0.119 + 0.229 + 0.004 + 0.238 = 0.621$$

Q.4 Venn Diagram on Probability

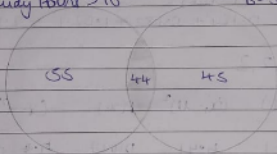
Set A = Students who study more than 10 hours in a week

Set B = students who attend more than 80% of classes

$P(A \cap B)$ = The one who study both condition

A = Study Hours > 10

B = Student > 80
Attendance



Venn Diagram of Student
Hours vs Attendance

Only A (Study > 10 hrs, attendance $< 80\%$)
55 students

Only B (Attendance $> 80\%$, Study ≤ 10 hrs)
45 students

Neigher A or B (SS + H + W) - 200 = 56 Students

Q5 Contingency table and probability

	Pass	Fail	Total
Gr D = Yes	90	30	120
Gr D = NO	50	30	80
Total	140	60	200

Joint probability: $P(\text{Gr D = Yes and Pass}) = 90/200 = 0.45$

Marginal probability: $P(\text{Pass}) = 0.70$

Conditional Probability: $P(\text{Pass} | \text{Gr D = Yes}) = 90/120 = 0.75$

Q6 Understanding Relationship

Intuition:

$$\rightarrow P(\text{Pass} | \text{Yes}) = \frac{P(\text{Pass and Yes})}{P(\text{Yes})}$$

mean:

$\rightarrow$ Out of only the students who took part in group discussion, what fraction of them passed?

$\rightarrow$ also we shrink our sample space from all 200 students down to just the 120 who participated, then ask what fraction of the smaller group passed. Standard given that we already know one thing

about the (Students), how does that update our belief about the other?

- Independent, Dependent or mutually exclusive?

→ $P(\text{Pass}) = 0.70$ (overall)

but,

$P(\text{Pass} | \text{Yes}) = 0.75$ and $P(\text{Pass} | \text{No}) = \frac{50}{80} = 0.625$

Since $P(\text{Pass} | \text{Yes}) \neq P(\text{Pass})$ knowing all Students joined group discussion changes the probability of passing so the two events are dependent. They are not mutually exclusive either. And joint probability $P(\text{Yes and Pass}) = 0.45$ is not zero - both events happen together for many Students.

0.7 Bayes Theorem Application

- Given $P(\text{High Attendance} | \text{Pass}) = 0.90$,
 $P(\text{High Attendance} | \text{Fail}) = 0.40$ $P(\text{High Attendance}) = 0.60$ we need $P(\text{Pass} | \text{High Attendance})$

Step 1 Find $P(\text{Pass})$ using the Law of total probability:

$$P(\text{HA}) = P(\text{Pass}) \cdot P(\text{Pass}) + P(\text{HA} | \text{fail}) \cdot P(\text{fail})$$

$$0.60 = 0.70 \cdot P(\text{Pass}) + 0.40 (1 - P(\text{Pass}))$$

$$0.60 = 0.40 + 0.30 P(\text{Pass}) \Rightarrow P(\text{Pass}) = 0.20/0.30 \\ = 0.667$$

Step 2 Apply Bayes theorem:

$$P(\text{Pass} | \text{HA}) = \frac{[P(\text{HA} | \text{Pass}) \cdot P(\text{Pass})]}{P(\text{HA})}$$

$$P(\text{Pass} | \text{HA}) = (0.70 \times 0.667) / 0.60 = 0.778 / 0.60 \\ = 0.778$$

Result: A student who has high attendance (280%) has about 77.8% probability of having passed the exam.